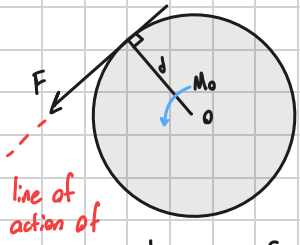
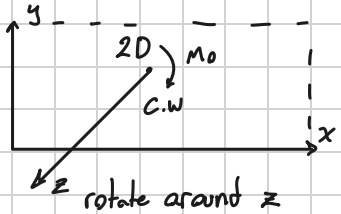
Week 7

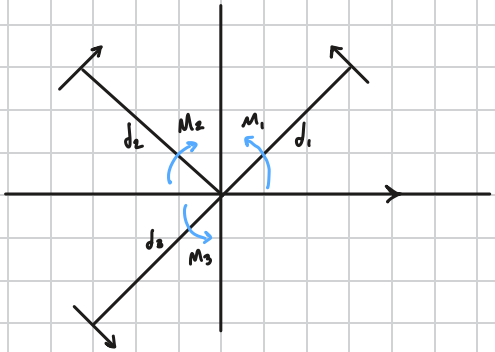
Compute moment of force in 2D

$$M_O = F \cdot d$$





direction of moment will be C.W or C.C.W

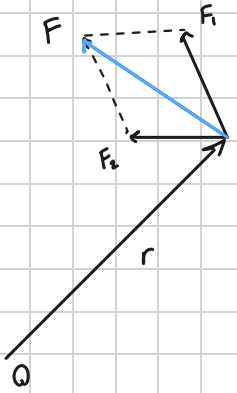


Scalor notation

$M_1 = F_1 d_1$

$M_2 = -F_2 d_2$

$M_3 = F_3 d_3$



$\vec{F} = \vec{F}_1 + \vec{F}_2$

$\vec{M}_O = \vec{r} \times \vec{F}$

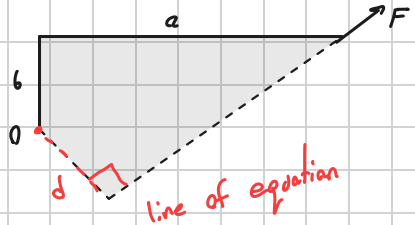
$\vec{M}_O = \vec{r} \times \vec{F}_1 + \vec{r} \times \vec{F}_2$

$= \vec{r} \times (\vec{F}_1 + \vec{F}_2)$

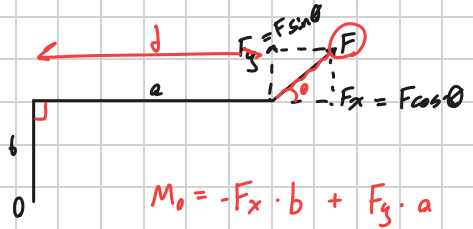
$= \vec{r} \times \vec{F}$

$M_o = F \cdot d$



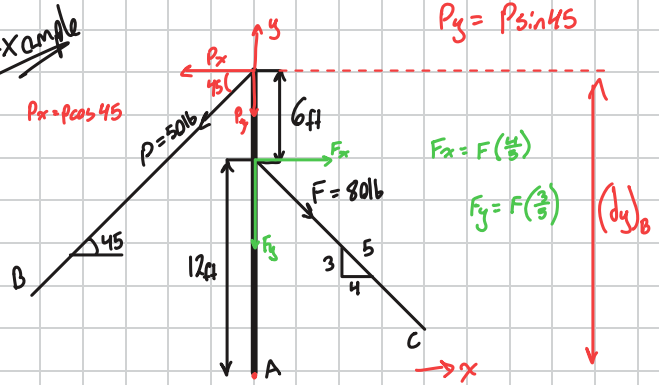


instead do this →



$M_o = -F_x \cdot b + F_y \cdot a$

Example



$P_y = P\sin 45$

$F_x = F(\frac{4}{5})$

$F_y = F(\frac{3}{5})$

$(d_y)_B$

$(M_B)_A = (P\cos 45)(12+6) + (P\sin 45)(0)$

$= 636$ lb·ft ↺ c.c.w

$(M_C)_A = -F_x(12) + F_y(0)$

$= -F(\frac{4}{5})(12)$

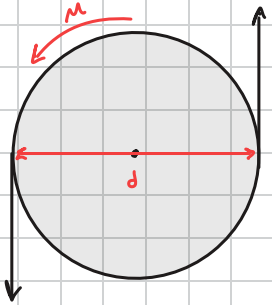
$= -786$ lb·ft ↺ c.c.w

$= 786$ lb·ft ↻ c.w

pole will rotate C.W

$(M_B)_A = ?$ $(M_C)_A = ?$

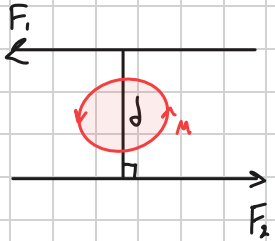
Moment of a couple



$M_C = F \cdot d$

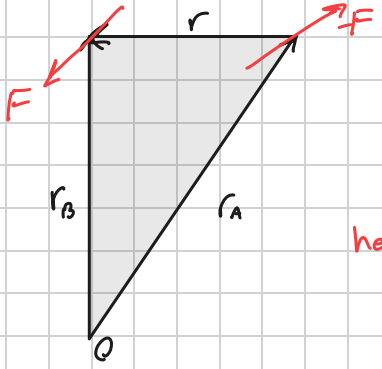
$\Sigma F_y = F - F$

$= 0$



$M_O = r \times F$ (Cross) → 3D

$M = F \cdot d$ → 2D



$r = r_B - r_A$

hence

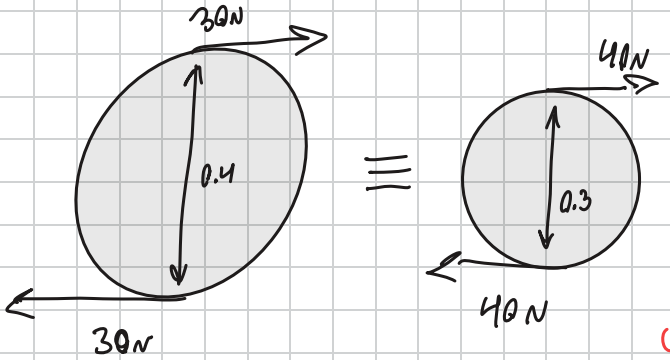
$M = r \times F$

$M_O = r_A \times (-F) + r_B \times F$

$= (r_B - r_A) \cdot F$

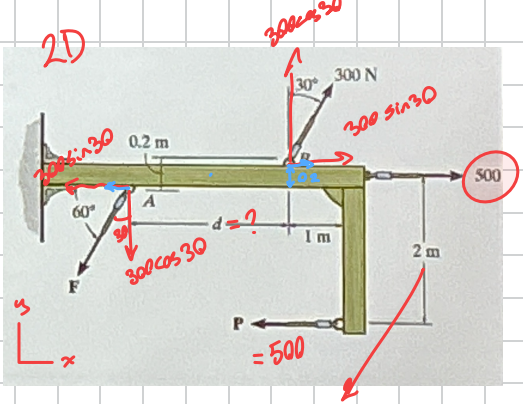
$M_C = \vec{r}_{AB} \times \vec{F}$

$M_C = \vec{r}_{BA} \times -\vec{F}$



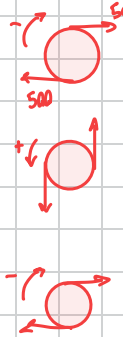
Multiple couples

$M_R = M_1 + M_2$



$P = 500N$

$F = 300N$



C.C.W $\circlearrowleft$ $M_C = -500(2) + 300\cos 30(d) - 300\sin 30(0.2)$

$M_C = 0 \Rightarrow$ find d

$0 = -1000 + 259.8d - 30$

$\therefore d = 3.96m$